

A negative answer to the symmetric-matrix TTO question

Literature-status note for arXiv:2306.12183

July 21, 2026

Status. Literature already answered.

1 Source question

O’Loughlin and Virtanen ask in Question 4.1 on page 12 of [1]:

Is every symmetric matrix unitarily equivalent to a direct sum of TTOs?

Here “symmetric” means complex symmetric as a matrix, and TTO abbreviates truncated Toeplitz operator. The question is motivated by the possibility of using TTOs as model operators for numerical ranges and Crouzeix’s conjecture.

2 Later negative answer

O’Loughlin’s July 2026 paper [2] answers this question negatively. Corollary 4.4 on page 7 states that for every $n \geq 10$, the question is false. More explicitly, for each such n there exists an irreducible complex symmetric $n \times n$ matrix that is not unitarily equivalent to a truncated Toeplitz operator.

The proof uses semialgebraic dimension. Let $\mathcal{CS}_n$ be the real vector space of complex symmetric $n \times n$ matrices and let $\mathcal{I}$ be its irreducible locus. The answering paper identifies a semialgebraic family $\mathcal{C}$ containing symmetric matrix representations of n -dimensional TTOs and proves

$$\dim_{\mathbb{R}} \mathcal{U}(\mathcal{C}) \leq 7n - 6 + \frac{n(n-1)}{2} < n(n+1) = \dim_{\mathbb{R}}(\mathcal{I} \cap \mathcal{CS}_n) \quad (n \geq 10).$$

Thus not every irreducible symmetric matrix has a TTO model. An irreducible matrix cannot be unitarily equivalent to a nontrivial direct sum, so this also rules out a direct sum of TTOs and answers the source question.

3 Scope

The result is existential: it proves that counterexamples form a dimension-theoretically unavoidable family but does not display a particular matrix. It settles the universal question negatively; the individual sizes $4 \leq n \leq 9$ remain open. The negative answer removes one proposed TTO-model route but does not disprove or prove Crouzeix’s conjecture.

References

- [1] R. O’Loughlin and J. Virtanen, *Crouzeix’s conjecture for classes of matrices*, arXiv:2306.12183v2, 2023.
- [2] R. O’Loughlin, *Semialgebraic Dimension and Truncated Toeplitz Models for Complex Symmetric Matrices*, arXiv:2607.14019v1, 2026.